Faculty of Computing

[CCN]

Lab No 7 Tasks

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Task 1:

Send a ping command for both HUB and SWITCH network using command prompt in packet tracer and notice the results in simulation mode.

Scenario:

Create simplest hub-based and switched-based networks, ideal for studying the different processes between hubs and switches. Add generic PCs and 2950-24 switches for one topology and generic hub with PCs the other topology.

Set the IP addresses ranging from 192.168.1.1 to 192.168.1.5 for one topology and 192.168.2.1 to 192.168.2.5 for another topology. All should have subnet masks 255.255.255.0.

Reflect:

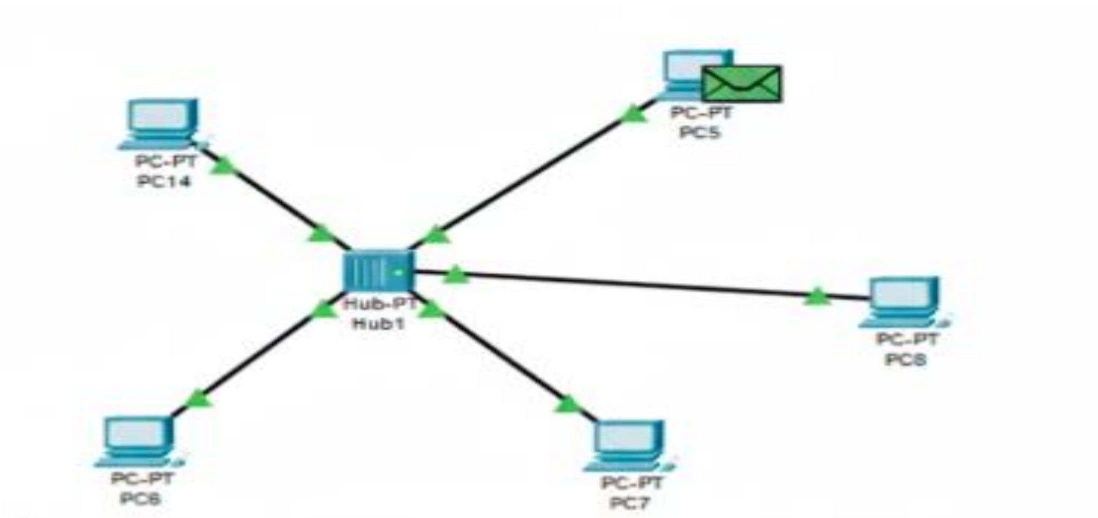
Go to simulation mode. In event list filters, enable only ICMP and ARP. Using the "Simple PDU", issue a ping from PC 0 to PC 1. Play the simulation. Pay close attention to how the hub processes the ICMP and ARP packets. After that, once again, use" Add simple PDU to issue a ping from PC1 to PC 0. Play the simulation again. How has the behavior of the hub changed from the first and second attempts, if at all?

Still in simulation mode, in the event List Filters, enable only ICMP and ARPO. Using the Add simple PDU issue a ping from PC 3 to PC 4. Play the simulation. Pay close attention to how the switch processes the ICMP and ARP packets. Now check for the behavior for the Switch.

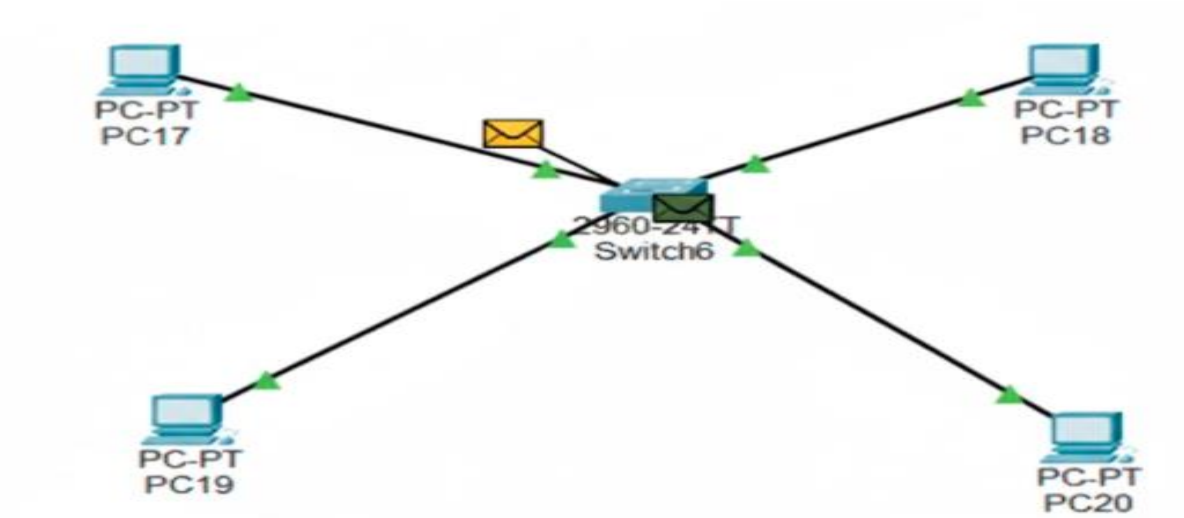
In what ways did the switch processes the packets similarly or differently from Hub between the first and second ping attempts?

Solution:

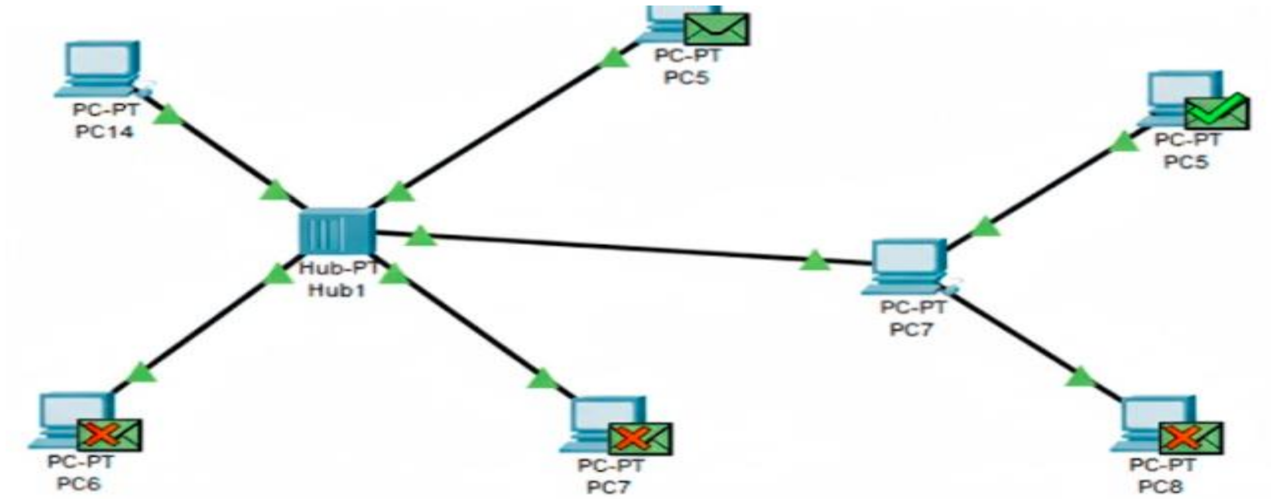
PC 5 send PDU back to HUB:



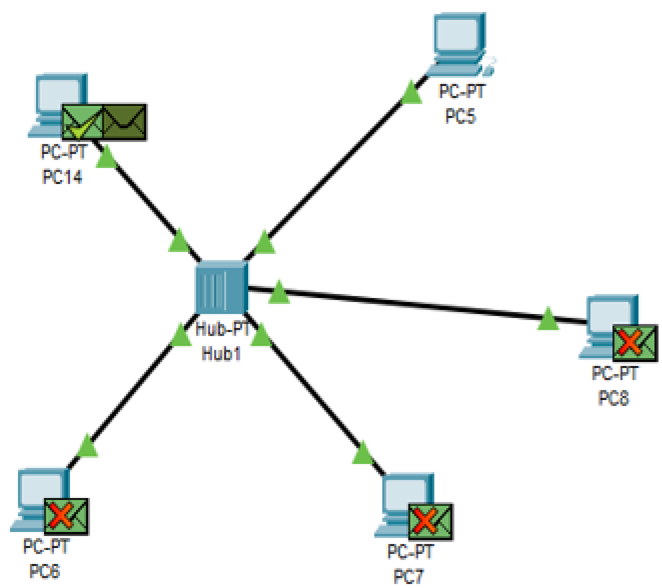
PDU goes to HUB from PC 14:



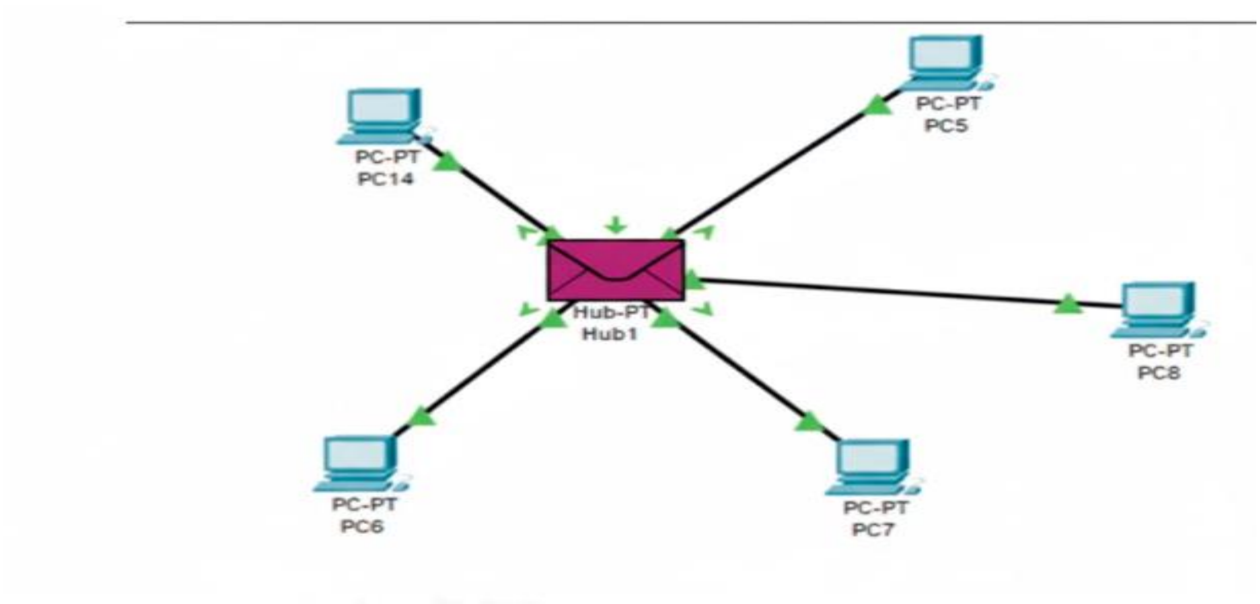
HUB broadcasts PDU to all PC's but only PC 5 accepts it as it is the destination:



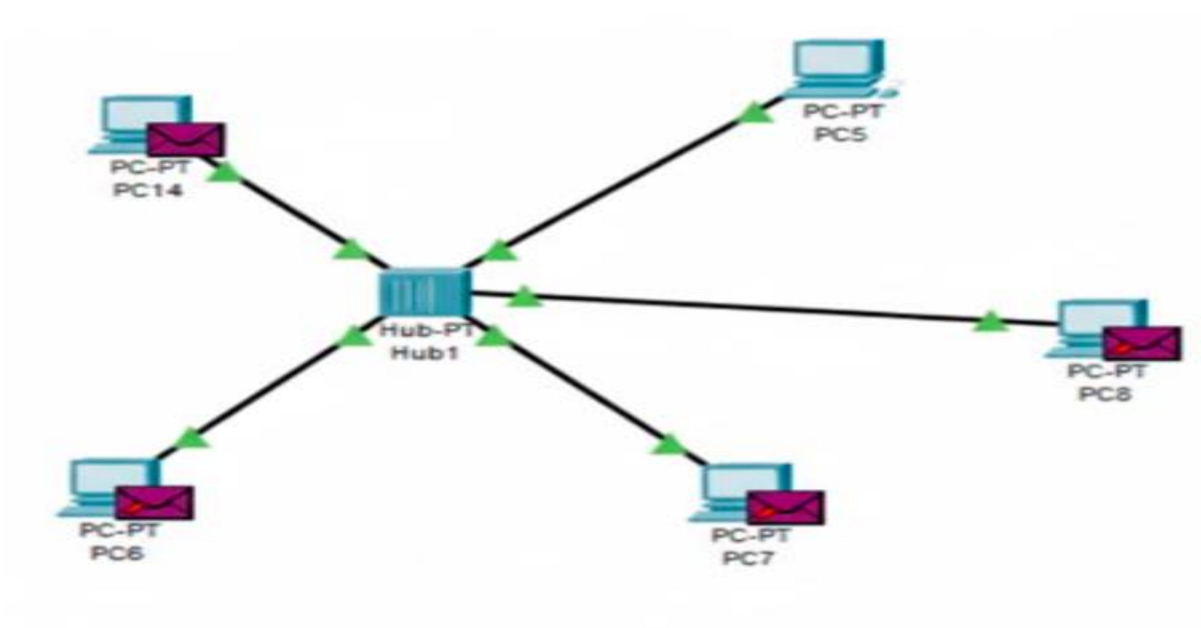
HUB again broadcasts PDU but only PC 14 accepts it.

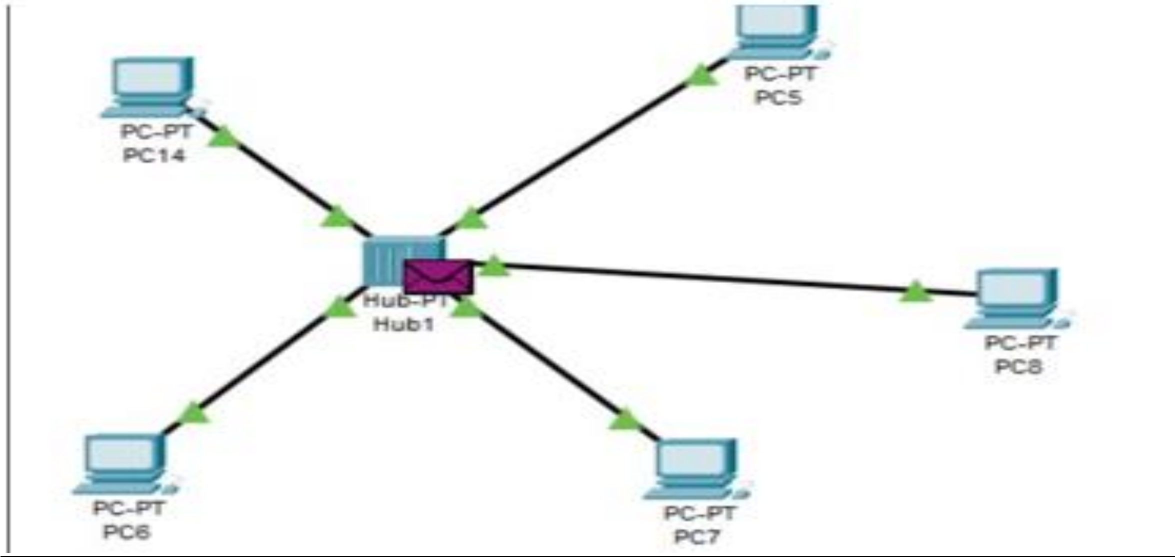


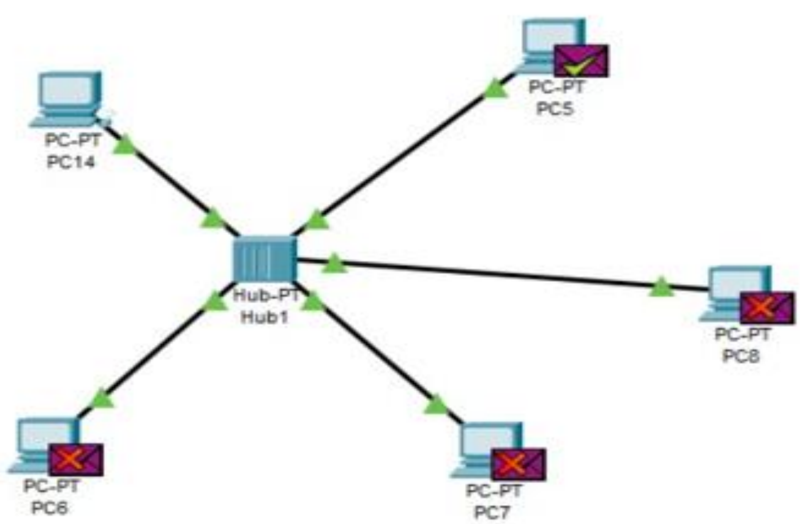
PDU from PC5 goes to HUB:



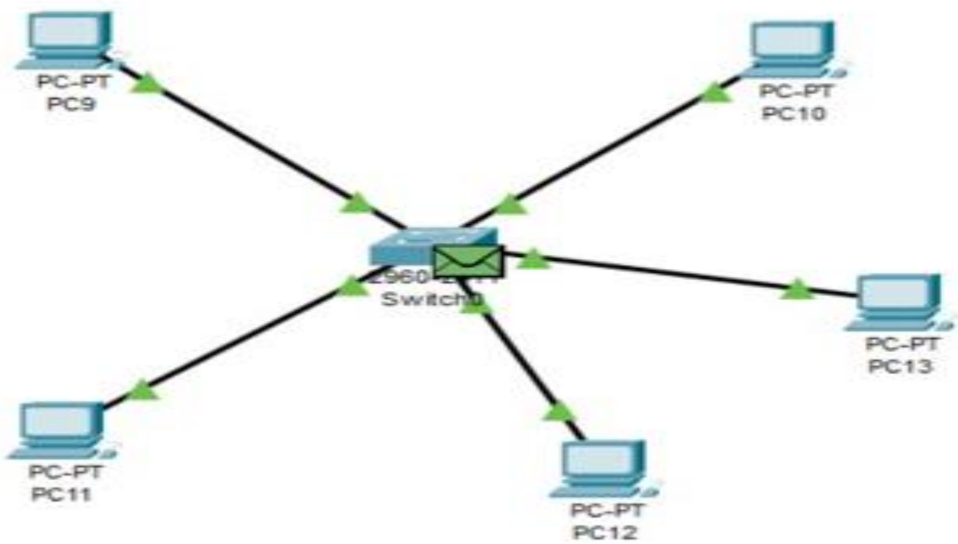
HUB broadcasts PDU to all PCs except PC5:

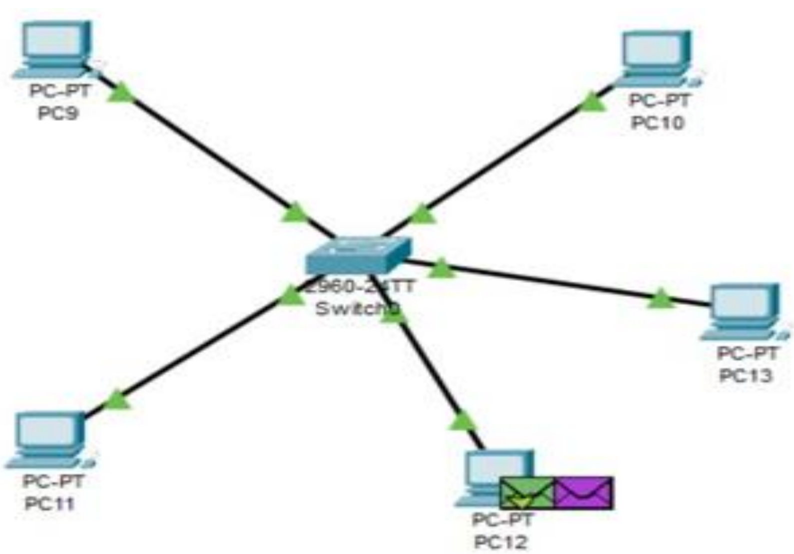



PC-PT
PC14
PC-PT
PC5
Hub-PT
Hub1
PC-PT
PC8
PC-PT
PC6
PC-PT
PC7


PC-PT
PC14
PC-PT
PC5
Hub-PT
Hub1
PC-PT
PC8
PC-PT
PC6
PC-PT
PC7

Switch:


PC-PT
PC9
PC-PT
PC10
Switch0
PC-PT
PC13
PC-PT
PC11
PC-PT
PC12


PC-PT
PC9
PC-PT
PC10
2960-24TT
Switch0
PC-PT
PC13
PC-PT
PC11
PC-PT
PC12

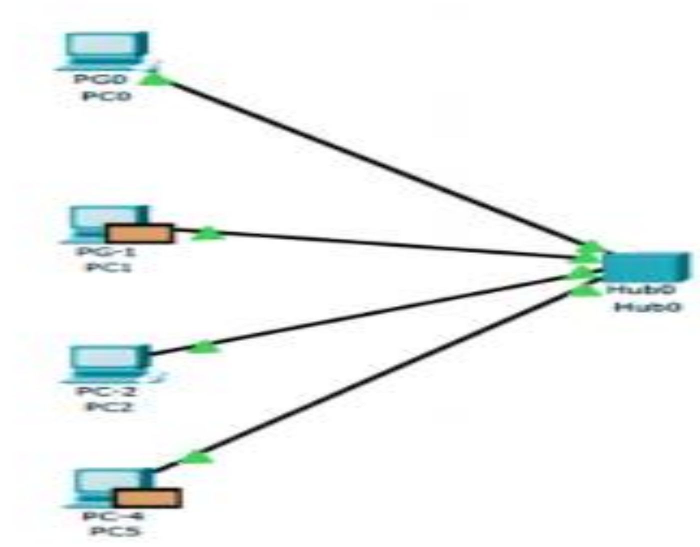
Task 2:

For the switched based network, set the new IP address and subnet mask for the PCs as follows.

PC	IP ADDRESSES	SUBNET MASK
PC0	192.168.1.36	255.255.255.224
PC1	192.168.1.37	255.255.255.224
PC2	192.168.1.38	255.255.255.224
PC3	192.168.1.91	255.255.255.224

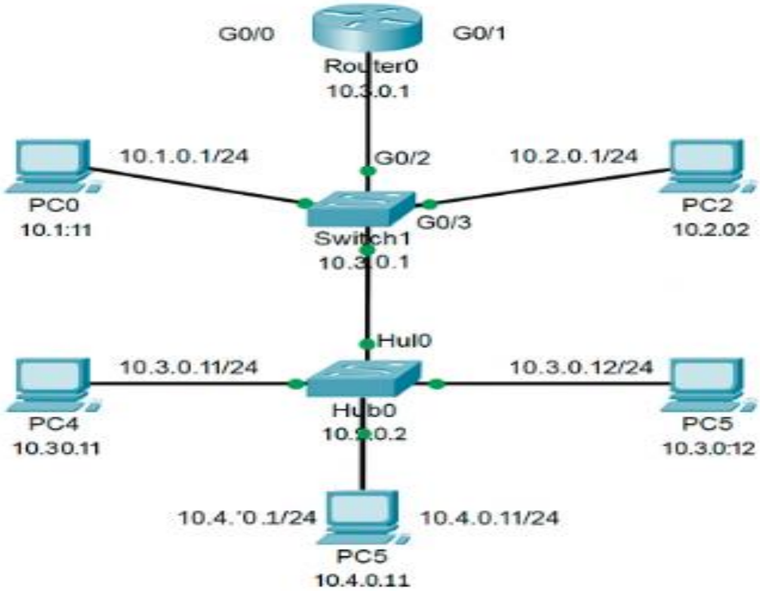
Tell about the communication. Communicated or not? If yes why.

Solution:

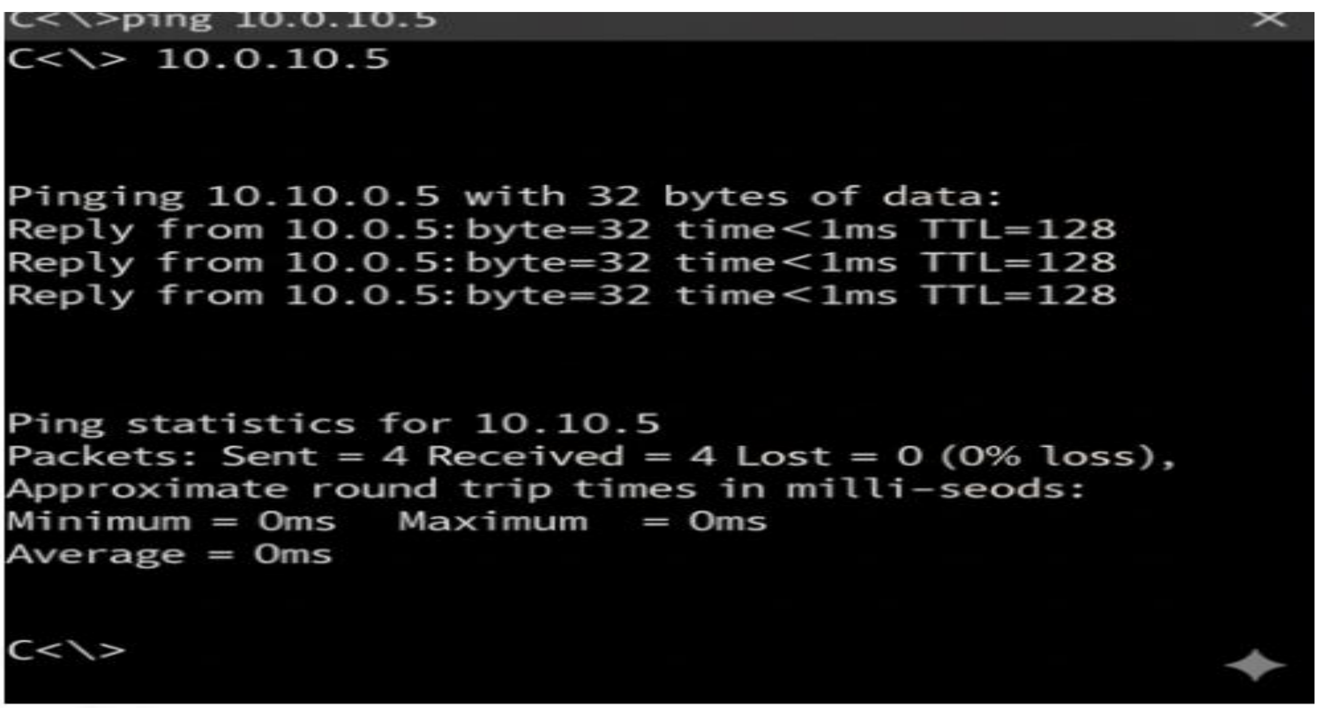


Task 3:

Create a star topology using a class A addressing scheme. Use Router to connect the end devices. Explain the behavior of the ping on each device. Use one router with 4 networks, 2 switches and 1 hub.



Pinging PC5 (10.0.20.2) and PC6 (10.0.20.3) within a network:



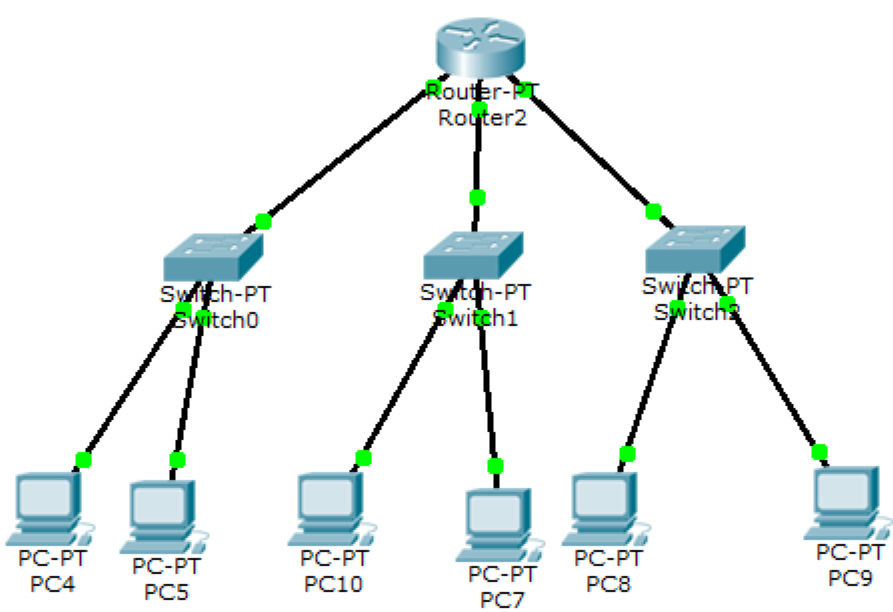
Pinging 10.0.30.2 and 10.0.40.2. 2different networks:

```
C\\Users\Admin>
Pinging
Pinging ing 10.0.400.2  with 32 bytes of data:
Reply from 10.0.440.2: byte=32 time< 1ms TTL=127
Reply from 10.0.440.2: byte=32 time< 1ms TTL=127
Reply from 10.0.440.2: byte=32 time< 1ms TTL=127


Ping statisics for 10.0.40.2:
Packets: Sent = 4, Received = 4 Lost = 0 (0% loss);
Approximate round trip times in milli-seonds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C\\Users\Admin>
```

Task 4:

Create the topology shown in the below diagram



Send packets and ping commands and provide results with explanation.

<u>Pinging PC5 (10.0.1.2) and PC6 (10.0.1.3). Within a same network:</u>

```
C:>ping 10.0.1.3

Pinging 10.0.1.3 with 32 bytes of data

Reply from 10.0.1.3: bytes=32 time=9ms
Reply from 10.0.1.3: bytes=32 time=4ms
Reply from 10.0.1.3: bytes=32 time=4ms
Reply from 10.0.1.3: bytes=32 time=4ms

Ping statistics for 10.0.1.3:
     Packets: Sent = 4, Received = 4,
Approximate round trip times in milli-
    Minimum = 4ms, Maximum = 9ms, Aver
C:>
```

Pinging PC6 (10.0.1.3) and PC7 (10.0.2.2). Within different network:

```
Cisco Packet Tracer PC Command Line 1.0
C:>>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:

Reply from 10.0.2.2: bytes=32 time<1ms TT
Reply from 10.0.2.2: bytes=32 time<1ms TT
Reply from 10.0.2.2: bytes=32 time<1ms TT
Reply from 10.0.2.2: bytes=32 time<1ms TT

Ping statistics for 10.0.2.2:
     Packets: Sent = 4, Received = 4, Lost
Approximate round trip times in milli-sec
   Minimum = 0ms, Maximum = 0ms, Average
C:\>
```